

2.13 Operators of Regular Expression

1. OR (+)
 2. Concatenation(.)
 3. Kleenestar (*)
 4. Kleenestar (+)
- $\left. \begin{array}{l} 1. \text{ OR (+)} \\ 2. \text{ Concatenation(.)} \end{array} \right\} \text{ Binary Operator}$
 $\left. \begin{array}{l} 3. \text{ Kleenestar (*)} \\ 4. \text{ Kleenestar (+)} \end{array} \right\} \text{ Unary Operator}$

Regular Expression	Equivalent Regular Set
$a + b$	$L(a + b) = \{a, b\}$
$a + a = a$	$L(a + a) = \{a\}$
$a + \phi = a$	$L(a + \phi) = \{a\}$
$a + \epsilon = \epsilon + a$	$L(a + \epsilon) = \{a, \epsilon\}$
$\epsilon + \epsilon = \epsilon$	$L(\epsilon + \epsilon) = \{\epsilon\}$
$\phi + \phi = \phi$	$L(\phi + \phi) = \{\}$
$a \cdot b = ab$	$L(a \cdot b) = \{ab\}$
$a \cdot \epsilon = a$	$L(a \cdot \epsilon) = \{a\}$
$\epsilon \cdot \epsilon = \epsilon$	$L(\epsilon \cdot \epsilon) = \{\epsilon\}$
$\phi \cdot \phi = \phi$	$L(\phi \cdot \phi) = \{\phi\}$
$a \cdot a = aa = a^2$	$L(a \cdot a) = aa = \{a^2\}$

2.14 Kleene star/ kleene closure / closure

R* (Kleene closure of R):

$$R^* = R^0 + R^1 + R^2 + R^3 + \dots = \epsilon + R + RR + RRR + \dots$$

Example:

$$a^* = \{a^0, a^1, a^2, a^3, \dots\} = \{\epsilon, a, aa, aaa, \dots\} = \text{Set of all strings over } a.$$

$$\phi^* = \phi^0 + \phi^1 + \phi^2 + \phi^3 + \dots = \epsilon + \phi + \phi + \phi + \dots = \epsilon + \phi = \epsilon$$

2.15 Positive Closure (Kleene Plus)

R⁺ (Positive Closure of R):

$$R^+ = R^1 + R^2 + R^3 + \dots$$

$$R^* = R^+ + R^0$$

Example:

- (1) $a^+ = \{a, aa, aaa, \dots\} = \{a^n \mid n \geq 1\}$
- (2) $\epsilon^+ = \epsilon = \epsilon^*$
- (3) $\phi^+ = \phi$

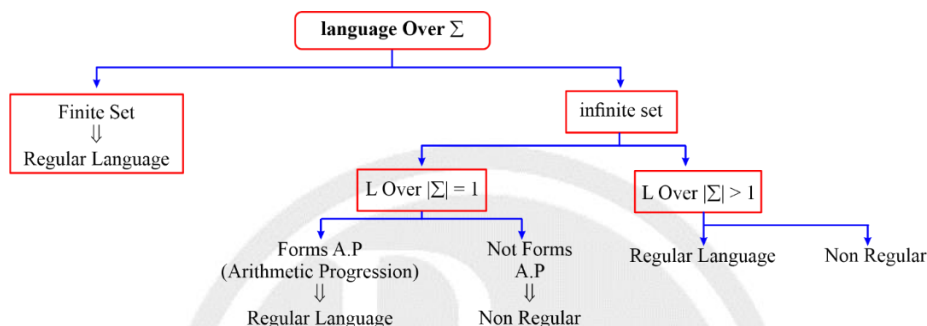
$R^* = R^{\geq 0}$ i.e. repeat R any number of time

$R^+ = R^{\geq 1}$ i.e. repeat R atleast 1 time.

Properties :

		OR	Concatenation
1.	Identity	ϕ	$\in$
2.	Associative	Yes	Yes
3.	Commutative	Yes	No
4.	Annihilator	Σ^*	ϕ

2.16 Identification of Regular Languages



- $L = \{a^n b^m \mid n, m \geq 0\} = a^* b^*$ (Regular language)
- $L = \{a^n b^m \mid m < n < 10\}$ (Finite Set, Regular language)
- $L = \{a^n b^m \mid m > n > 10\}$ (Non Regular language)
- $L = \{a^n b^m \mid m = n, m < 10\}$ (Regular language)
- $L = \{a^n b^m \mid m = n, m > 10\}$ (Non Regular language)
- $L = \{a^m b^n \mid \gcd(m, n) = 1\}$ (Non Regular language)
- $L = \{a^m b^n \mid \text{LCM}(m, n) = 1\}$ (Regular language)
- $L = \{a^n b^n \mid n \geq 0\}$ (Non Regular language)
- $L = \{a^n b^{2n} \mid n \geq 0\}$ (Non Regular language)
- $L = \{a^m b^n \mid m = \text{even}, n = \text{odd}\}$ (Regular language)
- $L = \{a^m b^n \mid m = n = \text{even}\}$ (Non Regular language)

OR

$$L = \{a^{2n} b^{2n} \mid n = \text{even}\}$$

- $L = \{a^*\}$ over $\Sigma = \{a\}$

= Regular language

- $L = \{a^{2n} \mid n \geq 0\}$ over $\Sigma = \{a\}$.

$$L = a^{2n} = (aa)^*$$

= Regular language

14. $L = \{a^{\text{Prime}}\}$ over $\Sigma = \{a\}$

= Non regular language

15. $L = \{a^{n^2} \mid n \geq 0\}$ over $\Sigma = \{a\}$

$L = \{\epsilon, a, aaaa, a^9, a^{16}, \dots\}$, FA Not possible for L. So, it's Non-Regular language.

16. $L = \{a^{2^n} \mid n > 0\}$ over $\Sigma = \{a\}$

Non Regular language

17. $L = \{a^{2^n} \mid n \leq 10\}$

= Regular language

18. $L = \{a^{n!} \mid n \geq 100\}$ over $S = \{a\}$

= Non Regular language

19. $L = \{a^{n^n} \mid n \geq 10\}$ over $\Sigma = \{a\}$

= Non Regular language

20. $L = \{a^{\text{Prime}}\}^*$ over $\Sigma = \{a\}$

$L = \text{complement of } \{a\} = \Sigma^* - \{a\} = \{\epsilon, a^2, a^3, a^4, a^5, a^6, a^7, \dots\} = \text{Regular language}$

21. $L = \{a^{\text{prime}} \mid \text{prime} < 100\}$ is finite language (regular)

22. $L = \{w \# w \mid w \in a^*\}$

$$L = \{a^n \# a^n\}$$


dependency

= Non Regular language

23. $L = \{w \# w \mid w \in (a+b)^*\}$

= Non Regular language

24. $L = \{w \# w^R \mid w \in a^*\}$ is non regular

$$L = \{a^n \# a^n\}$$


25. $L = \{w x w \mid w \in \{a, b\}^*, x \in \{a, b\}^+\}$

Put $w = \epsilon$, and $x = (a + b)^+$

$$L = (a + b)^+$$

= Regular language

26. $L = \{x w w \mid w \in \{a, b\}^*, x \in \{a, b\}^+\}$